

HP303S

PRECISION BAROMETER AND ALTIMETER SENSOR

Features

- Operation temperature: -40 ~ +85°C
- Supply voltage: 1.7V to 3.6V
- Pressure range: 300mBar ~ 1200mBar
- Pressure temperature sensitivity: <0.5Pa/K
- Pressure sensor resolution: $\pm 0.06\text{Pa}$
- Average current consumption: 1.7 μA for pressure measurement, 1.5 μA for temperature measurement at 1Hz sampling rate, 1x oversampling rate
- Standby: 0.5 μA
- 8-pin LGA, 2.0*2.5*0.95mm
- Stores up to 32 pressure or temperature measurements
- I2C and SPI (both with optional interrupt)



Figure1: 2.0*2.5*0.95mm

Applications

- Mobile Altimeter / GPS devices / tablet PCs
- Industrial Pressure and Temperature Sensor System
- Adventure and Sports watches
- Weather Station Equipment
- Indoor Navigation and Map Assist
- Data loggers for pressure, temperature and altitude
- Panel computer
- Barometers
- Gas meter

Descriptions

The HP303S is a miniaturized Digital Barometric Air Pressure Sensor with a high accuracy and a low current consumption, capable of measuring both pressure and temperature. The pressure sensor element is based on a capacitive principle which guarantees high precision during temperature changes. The small package makes the HP303S ideal for mobile applications and wearable devices.

The HP303S result FIFO can store up to 32 measurement results, allowing for a reduced host processor polling rate. Sensor measurements and calibration coefficients are available through the serial I2C or SPI interface. The measurement status is indicated by status bits or interrupts on the SDO pin.

1. Pin Configuration

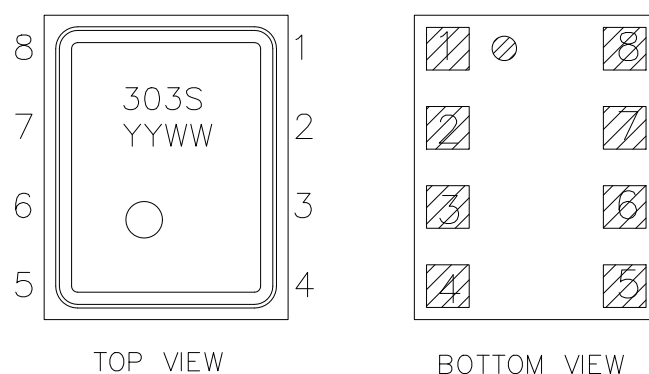


Figure2: Pin Configuration (bottom view)

Table1 Pin Descriptions

Pin	Name	SPI 3-wire	SPI 3-wire with interrupt	SPI 4-wire	I2C	I2C with interrupt
1	GND	Ground				
2	CSB	Chip select - active low	Chip select - active low	Chip select - active low	Not used - open (internal pull-up) or tie to VDDIO	Not used - open (internal pull- up) or tie to VDDIO
3	SDI	Serial data in/out	Serial data in/out	Serial data in	Serial data in/out	Serial data in/out
4	SCK	Serial Clock				
5	SDO	Not used	Interrupt	Serial data out	Least significant bit in the device address.	Interrupt pin and least significant bit in the device address.
6	VDDIO	Digital supply voltage for digital blocks and I/O interface				
7	GND	Ground				
8	VDD	Supply voltage for analog blocks				

2. Typical Application Circuit

Figure3: Application Circuit Example using the I2C serial interface.

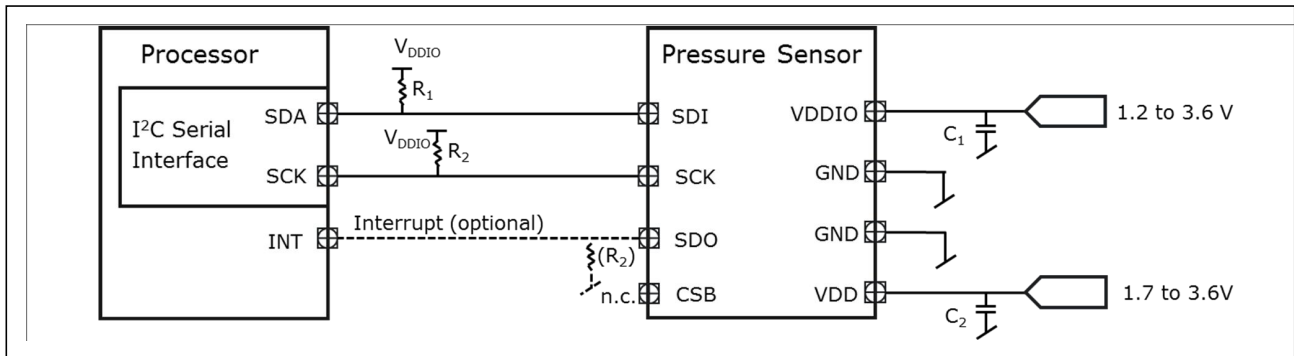


Figure4: Application Circuit Example using the SPI 4-wires serial interface

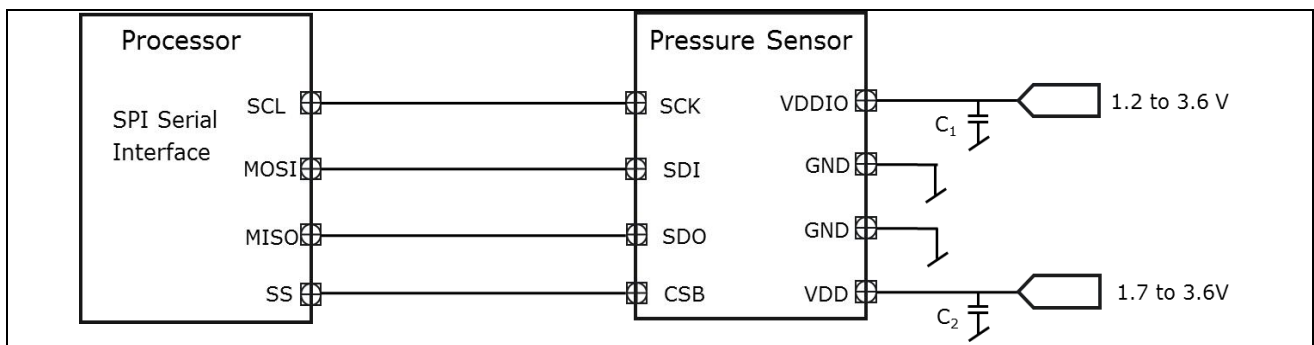


Figure5: Application Circuit Example using the SPI 3-wire serial interface

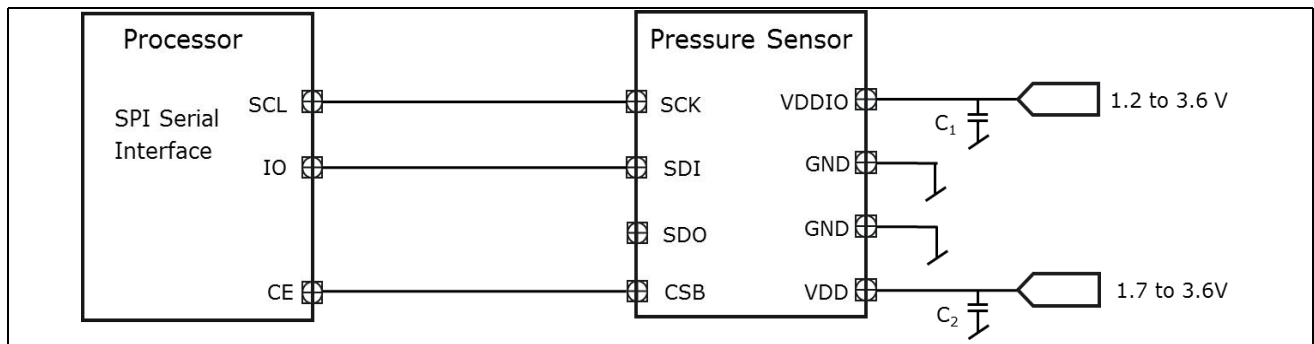


Table2 Component Values

Component	Symbol	Values			Unit	Note/Test Condition
		Min.	Typ.	Max.		
Pull-up/down Resistor	R1,R2	5		100	KΩ	R2 is optional and will set the address to 0x76 instead of 0x77
Supply Blocking Capacitor	C1,C2	100	100		nF	The blocking capacitors should be placed as close to the package pins as possible.

3. Mechanical and Electrical Specifications

3.1 Operating Range

Table3 Operating Range

Parameter	Symbol	Values			Unit	Note/Test Condition
		Min.	Typ.	Max.		
Pressure	Pa	300		1200	hPa	
Temperature	Ta	-40		85	°C	
Supply voltage	VDD	1.7		3.6	V	
Supply voltage IO	VDDIO	1.2		3.6	V	
Supply voltage ramp-up time	tvddup	0.001		5	ms	Time for supply voltage to reach 90% of final value.
Long term stability			0.5		hPa	12 months

3.2 Absolute Maximum Ratings

Maximum ratings are absolute ratings. Exceeding any one of these values may cause irreversible damage to the integrated circuit.

Table4 Absolute Maximum Ratings

Parameter	Symbol	Values			Unit	Note/Test Condition
		Min.	Typ.	Max.		
VDD and VDDIO	VDDxx_ma			3.6	V	
Voltage on any pin	Vmax			3.6	V	
Storage temperature	Ts	-40		125	°C	
Pressure	Pmax			10,000	hPa	
ESD	VESD_HB	-2		2	KV	HBM(JESD22-A114)

Note: Parameter above those listed as “Absolute Maximum Ratings” may cause permanent damage to the device. Exposure to maximum rating conditions for extended periods may affect device reliability.

3.3 Current Consumption

Test conditions (unless otherwise specified in the table): VDD=1.8V and VDDIO=1.8V. Typ. Values (PA=1000hPa and TA=25°C). Max./Min. values (PA=950-1050hPa and TA=0...+65°C).

Table5 Current Consumption

Parameter	Symbol	Values			Unit	Note/Test Condition
		Min.	Typ.	Max.		
Peak Current Consumption	I _{peak}		375		μA	During Pressure measurement
			295		μA	During Temperature measurement
Standby Current Consumption	I _{stb}		0.5		μA	
Current Consumption. (with 1 measurement per. second)	I _{1Hz}		3.3		μA	Low power
			13.5			Standard
			48.1			High precision

Note: The current consumption depends on both pressure measurement precision and rate. Please refer to the Pressure Configuration (PRS_CFG) register description for an overview of the current consumption in different combinations of measurement precision and rate.

3.4 Temperature Transfer Function

Test conditions (unless otherwise specified in the table): VDD=1.8V and VDDIO=1.8V. Typ. Values (PA=1000hPa and TA=25°C). Max./Min. values (PA=950-1050hPa and TA=0...+65°C).

Table6 Temperature Transfer Function

Parameter	Symbol	Values			Unit	Note/Test Condition
		Min.	Typ.	Max.		
Temperature accuracy	A _t		+/-0.5		°C	TA = 25°C
Temperature data resolution	A _{t_res}		0.01		°C	
Temperature measurement rate	f	1		200	Hz	

3.5 Pressure Transfer Function

Test conditions (unless otherwise specified in the table): VDD=1.8V and VDDIO=1.8V. Typ. Values (PA=1000hPa and TA=25°C).

Table7 Pressure Transfer Function

Parameter	Symbol	Values			Unit	Note/Test Condition
		Min.	Typ.	Max.		
Absolute pressure accuracy	A _{p_abs}		+/-100		Pa	PA=400-1050hPa TA=0...+65°C
Relative pressure accuracy	A _{p_rel}		+/-6		Pa	PA=950-1050hPa TA=25...+40°C
Pressure precision	A _{p_prc}		1.0		Pa RMS	Low Power
			0.5			Standard
			0.3			High Precision

Note: Pressure precision is measured as the average standard deviation. Please refer to the Pressure Configuration (PRS_CFG) register description for all precision mode options.

Power supply rejection	Ap_psr			0.063	Pa RMS	Measured with 217Hz square wave and broad band noise, 100mVpp
Pressure temperature sensitivity of calibrated measurements	Ap_tmp			0.5	Pa/K	1000hPa, 25...+40°C.
Pressure data resolution	Ap_res			0.06	Pa RMS	
Pressure measurement rate	f	1		200	Hz	
Pressure measurement time	t		5.2		ms	Low Power
			27.6			Standard
			104.4			High Precision

Note: The pressure measurement time (and thus the maximum rate) depends on the pressure measurement precision. Please refer to the Pressure Configuration (PRS_CFG) register description for an overview of the possible combinations of measurement precision and rate.

3.6 Timing Characteristics

Table8 Timing Characteristics

Parameter	Symbol	Values			Unit	Note/Test Condition
		Min.	Typ.	Max.		

Start-up timing

Time to sensor ready	TSensor_rdy			12	ms	The SENSOR_RDY bit in the Measurement Configuration register will be set when the sensor is ready.
Time to coefficients are available	TCoef_rdy			40	ms	The COEF_RDY bit in the Measurement Configuration register will be set when the coefficients can be readout.

Note: Start-up timing is measured from VDD>1.2V & VDDIO>0.6V or Soft Reset.

I2C Clock.	f_I2C			3.4	MHz	
SPI Clock	f_SPI			10	MHz	

4. Functional Description

4.1 Operating Modes

The HP303S supports 3 different modes of operation: Standby, Command, and Background mode.

Standby Mode

- Default mode after power on or reset. No measurements are performed.
- All registers and compensation coefficients are accessible.

Command Mode

- One temperature or pressure measurement is performed according to the selected precision.
- The sensor will return to Standby Mode when the measurement is finished, and the measurement result will be available in the data registers.

Background Mode

- Pressure and /or temperature measurements are performed continuously according to the selected measurement precision and rate. The temperature measurement is performed immediately after the pressure measurement.
- The FIFO can be used to store 32 measurement results and minimize the number of times the sensor must be accessed to read out the results.

Note: Operation mode and measurement type are set in the Sensor Operating Mode and Status (MEAS_CFG) register.

4.2 Mode transition diagram

The mode transition diagram is shown below.

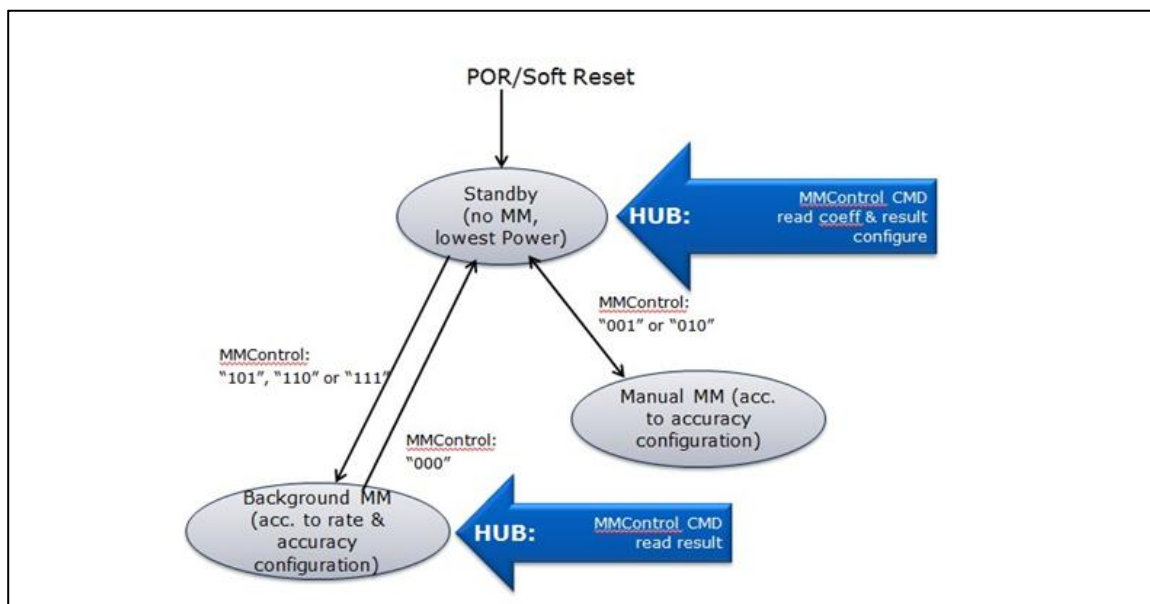


Figure6: Mode transition diagram

4.3 Start-up sequence

The start-up sequence of the HP303S is shown below. This diagram shows when the registers are accessible for read and /or write and also when the Pressure /Temperature measurements can start.

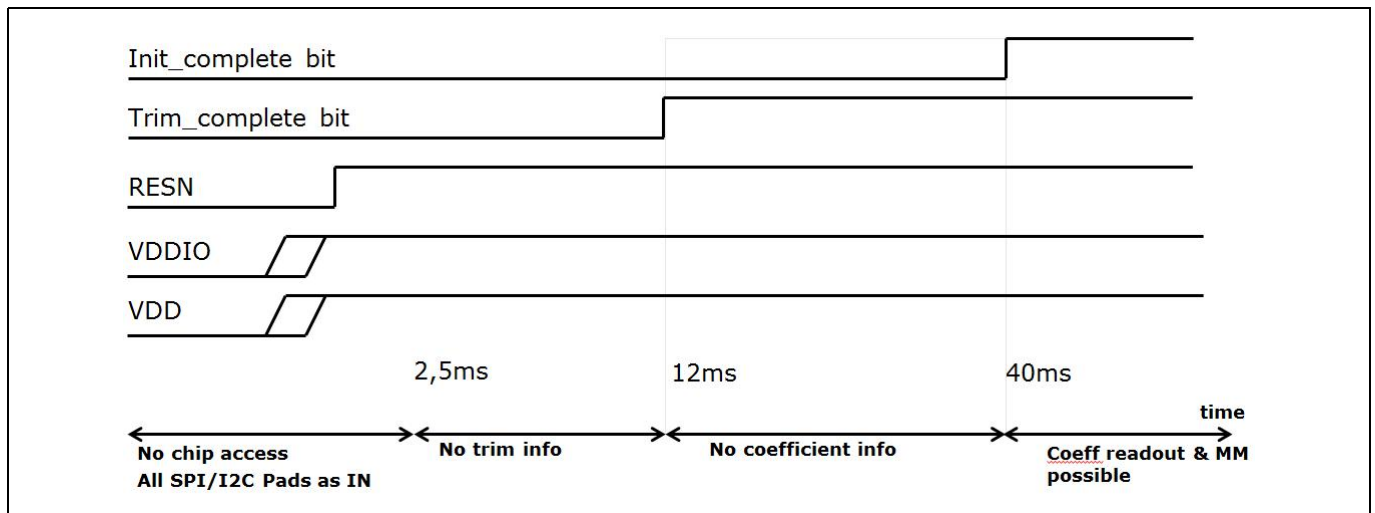


Figure7: Start-up sequence

4.4 Measurement Precision and Rate

Different applications require different measurement precision and measurement rates. Some applications, such as weather stations, require lower precision and measurement rates than for instance in door navigation and sports applications.

When the HP303S is in Background Mode, the measurement precision and rate can be configured to match the requirements of the application. This reduces current consumption of the sensor and the system.

In order to achieve a higher precision, the HP303S will read the sensor multiple times (oversampling), and combine the readings into one result. This increases the current consumption and also the measurement time, reducing the maximum possible measurement rate. It is necessary to balance the accuracy and data rate required for each application with the allowable current consumption.

The measurement precision, rate and time is set in the Pressure Configuration (PRS_CFG) and Temperature Configuration (TMP_CFG) registers. The register descriptions contain information about the current consumption and the possible combinations of measurement precision, time, and rate.

Enabling temperature measurements allows for compensation of temperature drift in the pressure measurement. The rates of these measurements can be set independently, but temperature compensation is more accurate when temperature and pressure measurements are taken together. This reduces the maximum pressure measurement rate, since: $\text{Rate}_{\text{temperature}} * \text{Time}_{\text{temperature}} + \text{Rate}_{\text{pressure}} * \text{Time}_{\text{pressure}} < 1 \text{ second}$. Measurement Settings and Use Case Examples contains a table with examples of combinations of pressure and temperature precision and rates for different use cases.

In the figure below is described the Temperature and Pressure measurements sequence in background mode.

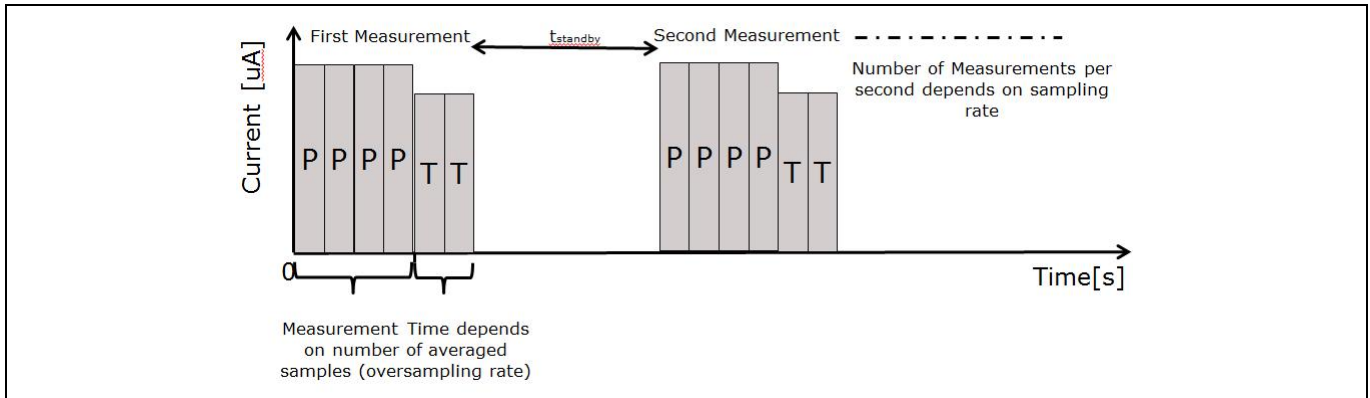


Figure8: Background mode temperature and pressure measurement sequence

4.5 Sensor Interface

The HP303S can be accessed as a slave device through mode '11' SPI 3-wire, SPI 4-wire, or I2C serial interface.

I2C interface

- The sensor's default interface.
- The sensor's address is 0x77 (default) or 0x76 (if the SDO pin is pulled-down to GND).

SPI interface

- The sensor will switch to SPI configuration if it detects an active low on the CSB pin. SPI 4-wire is the default SPI interface.
- To enable SPI 3-wire configuration, a bit must be set in the Interrupt and FIFO configuration (CFG_REG)

4.6 Register after startup

More details about digital interfaces are available in the Digital interfaces.

4.7 Interrupt

The HP303S can generate an interrupt when a new measurement result is available and /or when the FIFO is full. The sensor uses the SDO pin for the interrupt signal, and interrupt is therefore not supported if the interface is 4-wire SPI.

The interrupt is enabled and configured in the Interrupt and FIFO configuration (CFG_REG) register. In I2C configuration the SDO pin serves as both interrupt and as the least significant bit in the device address. If the SDO pin is pulled low the interrupt polarity must be set to active high and vice versa. The interrupt status can be read from the Interrupt Status (INT_STS) register.

4.8 Result Register Operation

After starting the measurements, the latest pressure and temperature raw data will be available in their respective result registers. Temperature measurement can be skipped. The temperature measurements can be disabled if there is a requirement to measure pressure rapidly, but it will make accurate temperature drift compensation impossible.

All measurement data can be read in a single command using auto-increment read. When FIFO is disabled, reading the result register will not affect the register value, it will only be updated when a new measurement is completed. When FIFO is enabled, the pressure result register will update to the next

value in the FIFO after each read. When all of the FIFO values have been read, the result register will be set to 0x800000.

4.9 FIFO Operation

The HP303S FIFO can store the last 32 measurements of pressure or temperature. This reduces the overall system power consumption as the host processor does not need to continuously poll data from the sensor but can go into standby mode for longer periods of time.

The FIFO can store any combination of pressure and temperature results, according to the background mode measurement rate settings. The pressure rate can for instance be set 4 times higher than the temperature rate and thus only every fifth result will be a temperature result. The measurement type can be seen in the result data. The sensor will set the least significant bit to:

- '1' if the result is a pressure measurement.
- '0' if it is a temperature measurement.

The sensor uses 24bits to store the measurement result. Because this is more bits than is needed to cover the full dynamic range of the pressure sensor, using the least significant bit to label the measurement type will not affect the precision of the result.

The FIFO can be enabled in the Interrupt and FIFO configuration register. The data from the FIFO is read out from the Pressure Data (PRS_Bn) registers regardless of whether the next result in the FIFO is a temperature or a pressure measurement.

When a measurement has been read out, the FIFO will auto increment and place the next result in the data register. A flag will be set in the FIFO Status register when the FIFO is empty and all following reads will return 0x800000.

If the FIFO is full, the FIFO_FULL bit in the FIFO Status (FIFO_STS) will be set. If the INT_FIFO bit in the Interrupt and FIFO configuration register (CFG_REG) is set, an interrupt will also be generated when the FIFO is full.

The FIFO will stop recording measurements results when it is full.

4.10 Calibration and Measurement Compensation

The HP303S is a calibrated sensor and contains calibration coefficients. These are used in the application (for instance by the host processor) to compensate the measurement results for sensor non-linearities. The sections that follow describe how to calculate the compensated results and convert the min to Pa.

How to Calculate Compensated Pressure Values

Steps:

1. Read the pressure calibration coefficients (c00, c10, c20, c30, c01, c11, and c21) from the Calibration Coefficient register

Note: The coefficients read from the coefficient register are 16bit 2's complement numbers.

2. Choose scaling factors kT (for temperature) and kP (for pressure) based on the chosen precision rate. The scaling factors are listed in Table 9.

3. Read the pressure and temperature result from the registers or FIFO.

Note: The measurements read from the result registers (or FIFO) are 24bit 2's complement numbers.

Depending on the chosen measurement rates, the temperature may not have been measured since the last pressure measurement.

4. Calculate scaled measurement results.

$$\text{Traw_sc} = \text{Traw}/\text{kT}$$

$$\text{Praw_sc} = \text{Praw}/\text{kP}$$

5. Calculate compensated measurement results.

$$\text{Pcomp (Pa)} = \text{c00} + \text{Praw_sc} * (\text{c10} + \text{Praw_sc} * (\text{c20} + \text{Praw_sc} * \text{c30})) + \text{Traw_sc} * \text{c01} + \text{Traw_sc} * \text{Praw_sc} * (\text{c11} + \text{Praw_sc} * \text{c21})$$

How to Calculate Compensated Temperature Values

Steps:

1. Read the temperature calibration coefficients (c0 and c1) from the Calibration Coefficients (COEF) register.

Note: The coefficients read from the coefficient register are 12 bit 2's complement numbers.

2. Choose scaling factor kT (for temperature) based on the chosen precision rate. The scaling factors are listed in Table 9.

3. Read the temperature result from the temperature register or FIFO.

Note: The temperature measurements read from the temperature result register (or FIFO) are 24bit 2's complement numbers.

4. Calculate scaled measurement results.

$$\text{Traw_sc} = \text{Traw}/\text{kT}$$

5. Calculate compensated measurement results.

$$\text{Tcomp (°C)} = \text{c0} * 0.5 + \text{c1} * \text{Traw_sc}$$

4.11 Compensation Scale Factors

Table9 Compensation Scale Factors

Oversampling Rate	Scale Factor (kP or kT)
1 (single)	524288
2 times (Low Power)	1572864
4 times	3670016
8 times	7864320
16 times (Standard)	253952
32 times	516096
64 times (High Precision)	1040384
128 times	2088960

4.12 Pressure and Temperature calculation flow

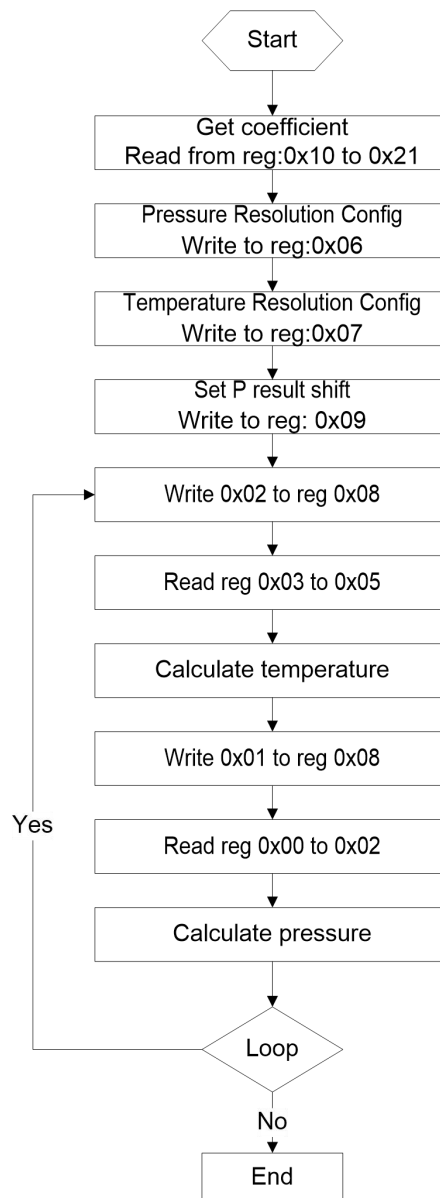


Figure9: flow chart

4.13 Digital interfaces

The HP303S measurement data, calibration coefficients, and configuration registers can be accessed through both the I2C and SPI serial interfaces.

The SPI interface can be configured to operate in 3-wire or 4-wire mode. In I2C and SPI 3-wire, an interrupt output can be implemented on the SDO pin. The SPI interface supports mode '11' only (CPOL=CPHA='1') in 4-wire and 3-wire configuration. The following commands are supported: single byte write, single byte read and multiple byte read using auto increment from a specified start address. The interface selection is done based on CSB pin status. If CSB is connected to VDDIO, the I2C interface is active. If CSB is low, the SPI interface is active. After the CSB has been pulled down once the I2C interface is disabled until the next power-on-reset.

4.13.1 I2C Interface

The I2C slave interface is compatible with Philips I2C Specification version 2.1. The I2C interface supports standard, fast and high speed mode.

The sensor's address is 0x77 (if SDO pin is left floating or pulled-up to VDDIO) or 0x76 (if the SDO pin is pulled- down to GND). The I2C interface uses the pins described in Table 1

The basic timing is shown in the diagram below:

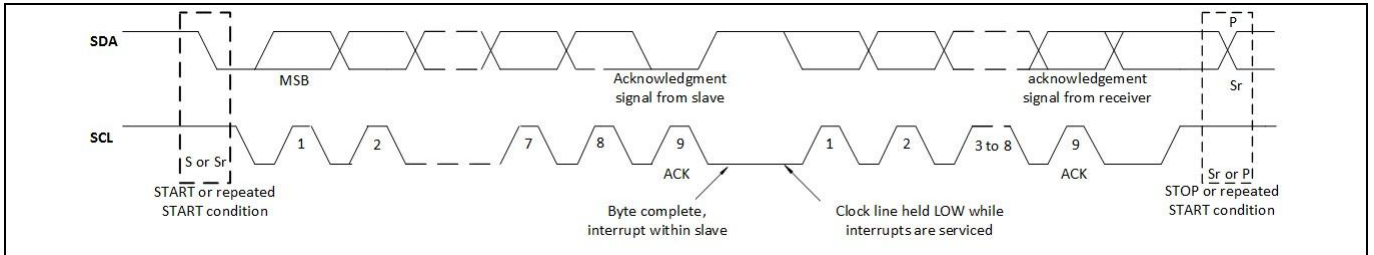


Figure10 I2Ctimingdiagram

In one access, without stop, incremental read (address is auto increment) and auto-incremental write is supported. The read and write access is described below:

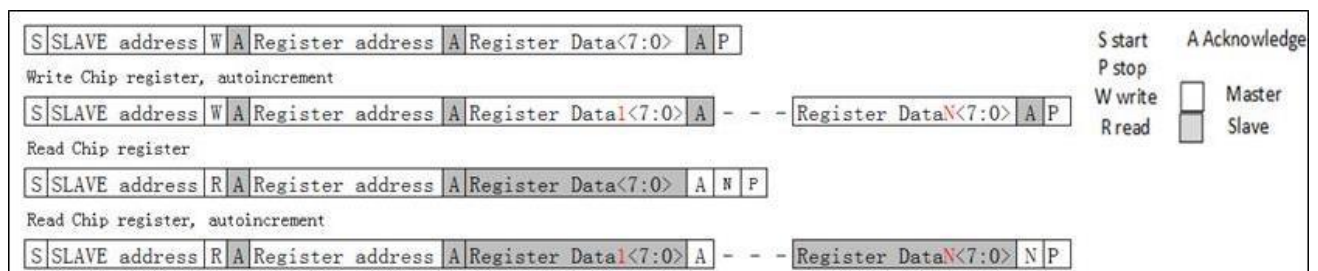


Figure11: I2C write and read commands

4.13.2 SPI Interface

The SPI interface is compatible with SPI mode '11' (CPOL = CPHA = '1'. The SPI interface has two modes: 4-wire and 3-wire.

The protocol is the same for both. The 3-wire mode is selected by setting '1' to the register Interrupt and FIFO configuration (CFG_REG)

The SPI interface uses the pins like in the Table 1 Refer to Application Circuit Example for connections instructions, The SPI protocol is shown in the diagram below

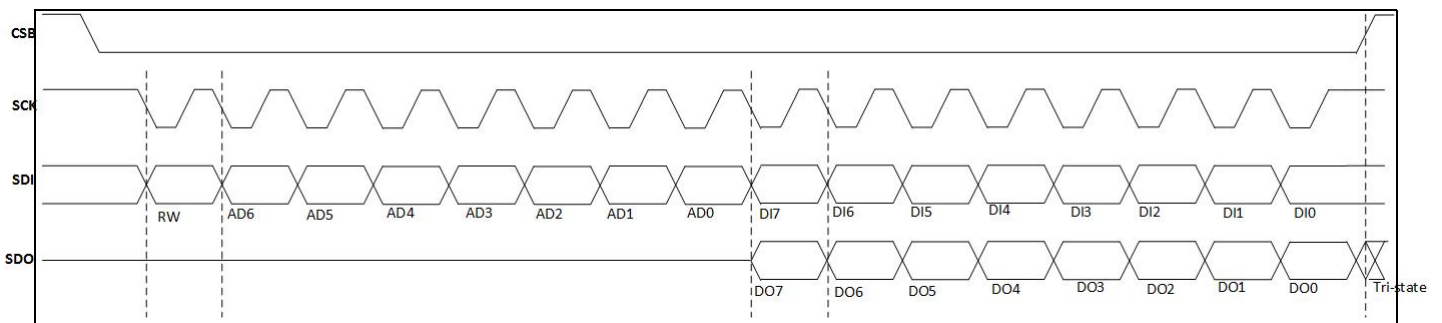


Figure12: SPI protocol 4-wire without interrupt

A SPI write is carried out by setting CSB low and sending a control byte followed by register data. The control byte consist of the SPI register address (full register address without bit 7) and the write command (bit7 = RW = '0'). Setting CSB high ends the transaction. The SPI write protocol is described in the diagram below.

A SPI read is initiated by setting CSB low and sending a single control byte. The control byte consist of the SPI

Register address (= full register address without bit7) and the read command (bit7 = RW = '1'). After writing the control byte, data is sent out of the SDO pin (SDI in 3-wire mode); the register address is automatic ally incremented. Sending CSB high ends the SPI read transaction. The SPI read protocol is shown in the diagram below:

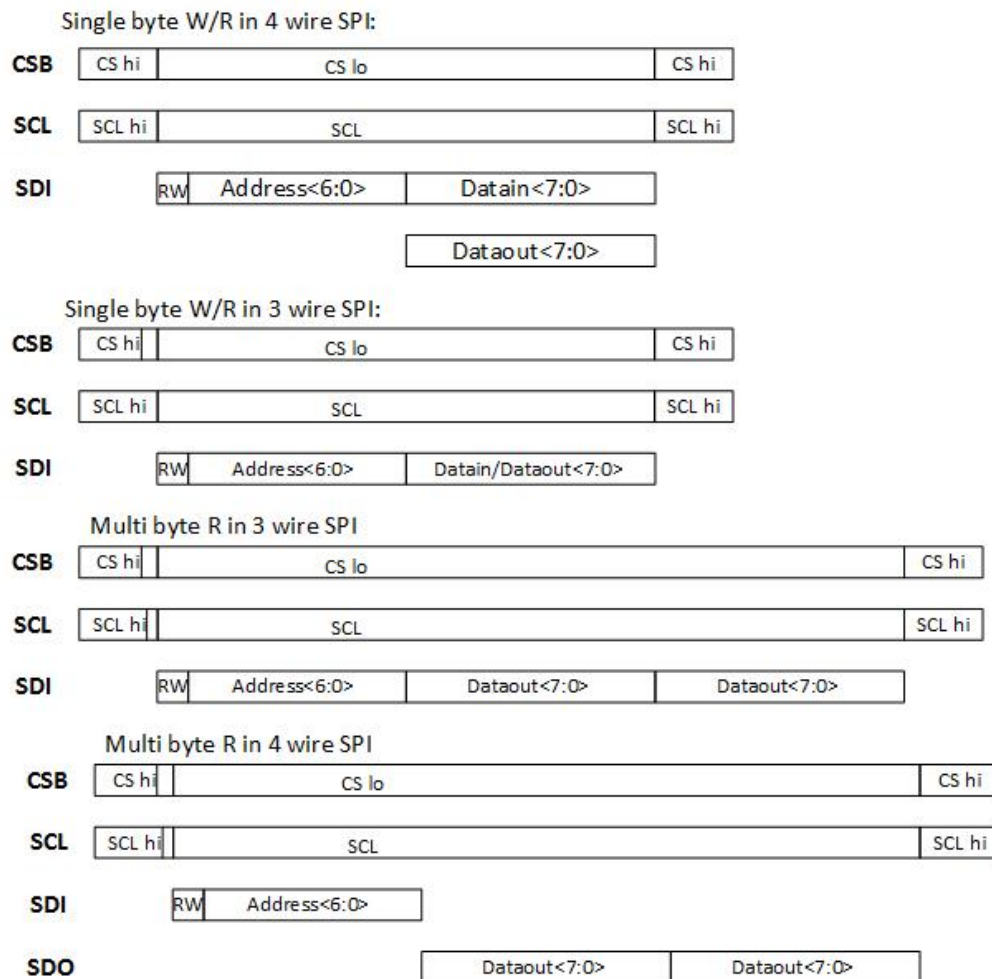


Figure13: SPI write, read protocol diagrams

4.13.3 Interface parameters specification

The general interface parameters are given in the table below:

Table10 Interface parameters

Parameter	Symbol	Values			Unit	Note/Test Condition
		Min.	Typ.	Max.		
Input voltage for low logic level at input pins	Vlow_in			0.3* VDDIO	V	VDDIO=1.2Vto3.6V
Input voltage for high logic level at input pins	Vhigh_in	0.7* VDDIO			V	VDDIO=1.2Vto3.6V
Output-low level for I2C	Vlow_SDI			0.1* VDDIO	V	VDDIO=1.8V, iol=2mA
Output voltage for low level at pin SDI for I2C	Vlow_SDI_1.2			0.2* VDDIO	V	VDDIO=1.20V, iol=1.3mA
Output voltage for high level at pins SDO, SDI	Vhigh_out	0.8* VDDIO			V	VDDIO=1.8V, iol=1mA(SDO, SDI)
Output voltage for high level at pins SDO, SDI	Vhigh_out_1.2	0.6* VDDIO			V	VDDIO=1.2V, iol=1mA(SDO, SDI)
Pull-up resistor	Rpull	60	120	180	kohm	Internal pull-up resistance to VDDIO
I2C bus load capacitor	Cb			400	pF	On SDI and SCK

4.13.4 I2C timings

The I2C timing is shown in the diagram below and corresponding values are given in the table below. The naming refers to I2C Specification version 2.1, the abbreviations used "S&F mode" = standard and fast mode, "HS mode" = high speed mode, Cb = bus capacitance on SDA line.

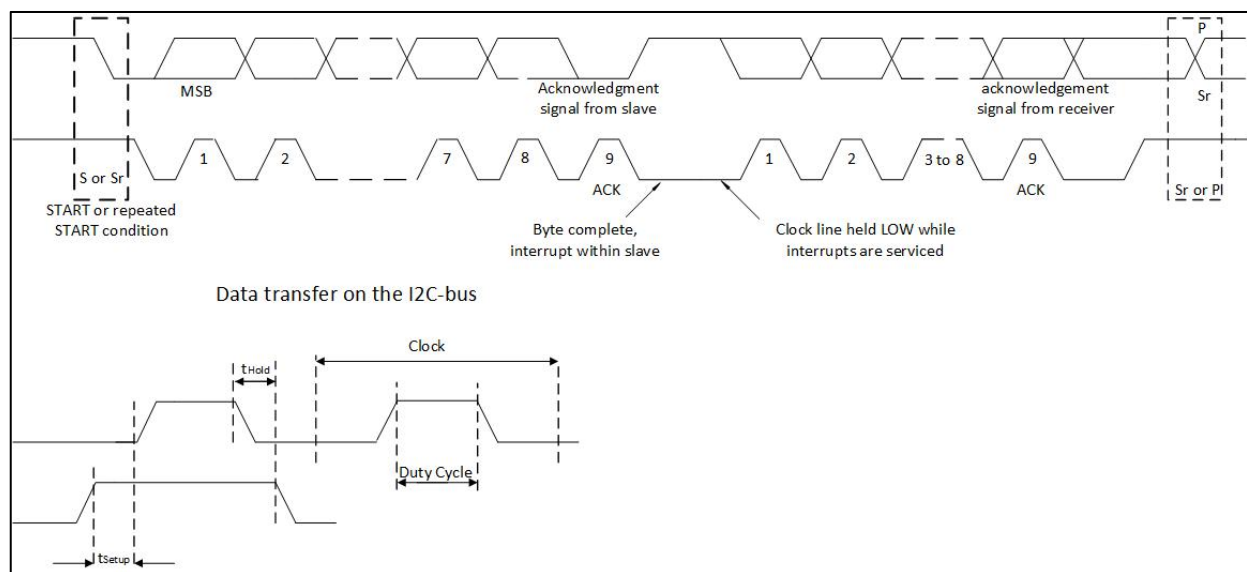


Figure14: I2C timing diagram

Table11 I2C timings

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
Data setup time on SDI pin	tSetup	20			ns	S&F mode
		5			ns	HS mode
Data hold time on SDI pin	tHold	0			ns	S&F&HS mode,
Duty Cycle	DC			70	%	S&F mode,
				55	%	HS mode,

4.13.5 SPI timings

The SPI timing diagram is shown in the figure below and the corresponding values are given in the table below. All timings apply both to 4-wire and 3-wire SPI.

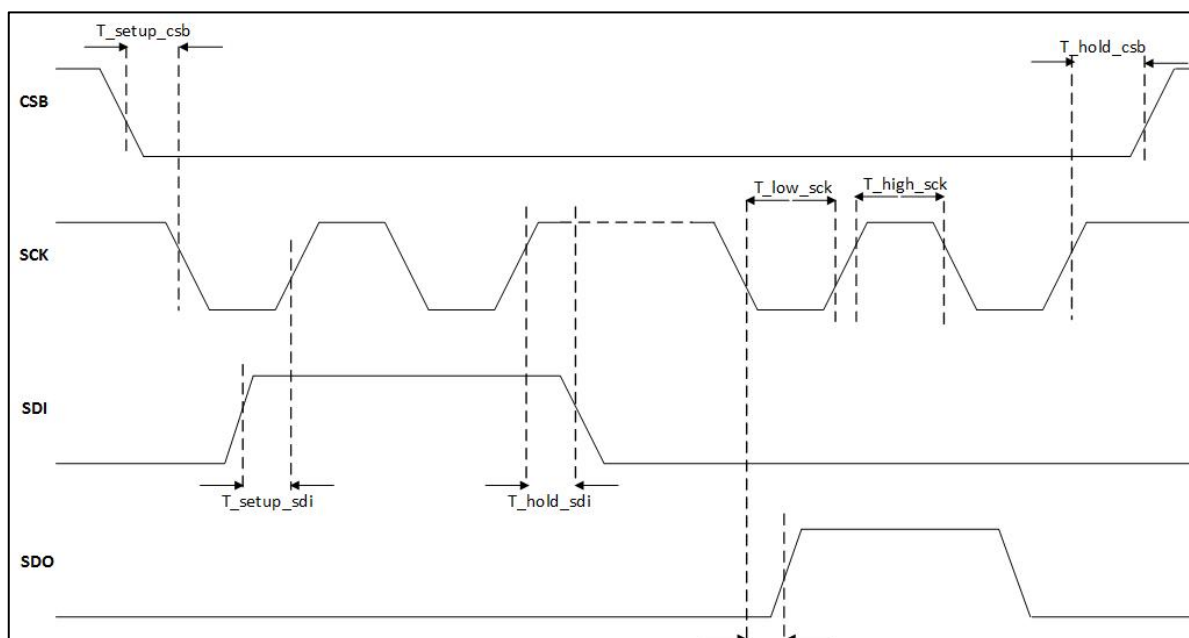


Figure15: SPI timing diagram

Table12 SPI timings

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
Duty Cycle(Thigh%)	SPI_DC	30			%	VDDIO=1.2V
		20			%	VDDIO=1.8V/3.6V
SDI setup time	T_setup_sdi	2			ns	
SDI hold time	T_hold_sdi	2			ns	
Clock	SPI_CLK			10	MHz	
CSB setup time	T_setup_csb	15			ns	
CSB hold time		15			ns	

5. Register Map

Table13 Register Map

Register Name	Addr.	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	Reset State
PSR_B2	0x00	PSR[23:16](r)								00h
PSR_B1	0x01	PSR[15:8](r)								00h
PSR_B0	0x02	PSR[7:0](r)								00h
TMP_B2	0x03	TMP[23:16](r)								00h
TMP_B1	0x04	TMP[15:8](r)								00h
TMP_B0	0x05	TMP[7:0](r)								00h
PRS_CFG	0x06	PM_RATE[7:4](rw)				PM_PRC[3:0](rw)				00h
TMP_CFG	0x07	TMP_RATE[7:4](rw)				TM_PRC[3:0](rw)				00h
MEAS_CFG	0x08	COEF_RDY(r)	SENSOR_RDY(r)	TMP_RDY(r)	PRS_RDY(r)	TMP_EXT(rw)	MEAS_CRTL[2:0](rw)			00h
CFG_REG	0x09	INT_HL(rw)	INT_SEL[2:0](rw)			TMP_SHIFT_EN(rw)	PRS_SHIFT_EN(rw)	FIFO_EN(rw)	SPI_MODE(rw)	00h
INT_STS	0x0A	-	-	-	-	-	INT_FIFO_FULL(r)	INT_TMP(r)	INT_PRS(r)	00h
FIFO_STS	0x0B	-	-	-	-	-	-	FIFO_FULL(r)	FIFO_EMPTY(r)	00h
RESET	0x0C	FIFO_FLUSH(w)	-	-	-	SOFT_RST[3:0](w)				00h
COEF	0x10-0x21	<see register description>								xxh
Reserved	0x22-0x2A	Reserved								xxh
COEF_SRC E	0x2B	TMP_COEF_SRCE(r)	Reserved							xxh

6. Register description

6.1 Pressure Data (PRS_Bn)

The Pressure Data registers contains the 24bit (3 bytes) 2's complement pressure measurement value. If the FIFO is enabled, the register will contain the FIFO pressure and/or temperature results (please see FIFO operation). Otherwise, the register contains the pressure measurement results and will not be cleared after read.

PRS_B2

The highest byte of the three bytes measured pressure value.

PRS_B2				Address:				00H			
Pressure (MSB data)				Reset value:				00H			
7	6	5	4	3	2	1	0				
PRS23	PRS22	PRS21	PRS20	PRS19	PRS18	PRS17	PRS16				
r											

Field	Bits	Type	Description
PRS[23:16]	7:0	r	MSB of 24bit 2's complement pressure data.

PRS_B1

The middle byte of the three bytes measured pressure value.

PRS_B1	Address:		01H				
Pressure (LSB data)	Reset value:		00H				
7	6	5	4	3	2	1	0
PRS15	PRS14	PRS13	PRS12	PRS11	PRS10	PRS9	PRS8
r							

Field	Bits	Type	Description
PRS[15:8]	7:0	r	LSB of 24bit 2's complement pressure data.

PRS_B0

The lowest byte of the three bytes measured pressure value.

PRS_B0				Address:				02H			
Pressure (XLSB data)				Reset value:				00H			
7	6	5	4	3	2	1	0				
PRS7	PRS6	PRS5	PRS4	PRS3	PRS2	PRS1	PRS0				
r											

Field	Bits	Type	Description
PRS[7:0]	7:0	r	XLSB of 24bit 2's complement pressure data.

6.2 Temperature Data (TMP_Tn)

The Temperature Data registers contain the 24 bit (3 bytes) 2's complement temperature measurement value (Unless the FIFO is enabled, please see FIFO operation) and will not be cleared after the read.

TMP_B2

The highest byte of the three bytes measured temperature value.

TMP_B2	Address:		03H				
Temperature (MSB data)	Reset value:		00H				
7	6	5	4	3	2	1	0
TMP23	TMP22	TMP21	TMP20	TMP19	TMP18	TMP17	TMP16
r							

Field	Bits	Type	Description
TMP[23:16]	7:0	r	MSB of 24bit 2's complement temperature data.

TMP_B1

The middle byte of the three bytes measured temperature value.

TMP_B1	Address:		04H				
Temperature (LSB data)	Reset value:		00H				
7	6	5	4	3	2	1	0
TMP15	TMP14	TMP13	TMP12	TMP11	TMP10	TMP9	TMP8
r							

Field	Bits	Type	Description
TMP[15:8]	7:0	r	LSB of 24bit 2's complement temperature data.

TMP_B0

The lowest part of the three bytes measured temperature value.

TMP_B0	Address:	05H					
Temperature (XLSB data)	Reset value:	00H					
7	6	5	4	3	2	1	0
TMP7	TMP6	TMP5	TMP4	TMP3	TMP2	TMP1	TMP0
r							

Field	Bits	Type	Description
TMP[7:0]	7:0	r	XLSB of 24bit 2's complement temperature data.

6.3 Pressure Configuration (PRS_CFG)

Configuration of pressure measurement rate (PM_RATE) and resolution (PM_PRC)

PRS_CFG Address: 06H

Pressure measurement configuration Reset value: 00H

7	6	5	4	3	2	1	0
PM_RATE[3:0]				PM_PRC[3:0]			
rw				rw			

Field	Bits	Type	Description
PM_RATE[3:0]	7:4	rw	Pressure measurement rate: 0000-1 measurements pr.sec. 0001-2 measurements pr.sec. 0010-4 measurements pr.sec. 0011-8 measurements pr.sec. 0100-16 measurements pr.sec. 0101-32 measurements pr.sec. 0110-64 measurements pr.sec. 0111-128 measurements pr.sec. 1000-25/16 measurements pr.sec. 1001-25/8 measurements pr.sec. 1010-25/4 measurements pr.sec. 1011-25/2 measurements pr.sec. 1100-25/1 measurements pr.sec. 1101-50 measurements pr.sec. 1110-100 measurements pr.sec. 1111-200 measurements pr.sec. Applicable for measurements in Background mode only.
PM_PRC[3:0]	3:0	rw	Pressure oversampling rate: 0000-Single (Low Precision). 0001-2 times (Low Power). 0010-4 times. 0011-8 times. 0100*)-16 times (Standard). 0101*)-32 times. 0110*)-64 times (High Precision). 0111*)-128 times. 1xxx*)-Reserved.

*Note: Use in combination with a bit shift. See Interrupt and FIFO configuration (CFG_REG) register.

Table14 Pressure measurement time (ms) and precision (Pa RMS) versus oversampling rate

Oversampling (PRC[3:0])	Single (0000)	2 times (0001)	4 times (0010)	8 times (0011)	16 times (0100)	32 times (0101)	64 times (0110)	128 times (0111)
Measurement time (ms)	3.6	5.2	8.4	14.8	27.6	53.2	104.4	206.8
Precision (Pa RMS)	2.0	1.0	0.7	0.6	0.5	0.4	0.3	0.3

Table15 Estimated current consumption (uA)

Oversampling (PRC[3:0])	Single (0000)	2 times (0001)	4 times (0010)	8 times (0011)	16 times (0100)	32 times (0101)	64 times (0110)	128 times (0111)
Measurements pr sec. (PM_RATE([2:0])								
1(0000)	2.6	3.3	4.8	7.7	13.5	25.0	48.1	94.3
2(0001)								
4(0010)								
8(0011)	Note: The current consumption can be calculated as the Measurement Rate * Current Consumption of 1 measurement per. sec.							n.a.
16(0100)							n.a.	n.a.
32(0101)						n.a.	n.a.	n.a.
64(0110)					n.a.	n.a.	n.a.	n.a.
128(0111)			n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
200(1111)		n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Note: The table shows the possible combinations of Pressure Measurement Rate and Oversampling when no temperature measurements are performed. When temperature measurements are performed the possible combinations are limited to Rate temperature x Measurement Time temperature + Rate pressure x Measurement Time pressure < 1 second.

6.4 Temperature Configuration (TMP_CFG)

Configuration of temperature measurement rate (TMP_RATE) and resolution (TMP_PRC)

TMP_CFG				Address:				07H							
Temperature measurement configuration				Reset value:				00H							
7		6		5		4		3		2		1		0	
TMP_RATE[3:0]								TMP_PRC[3:0]							
rw								rw							

Field	Bits	Type	Description
TMP_RATE[3:0]	7:4	rw	Temperature measurement rate: 0000-1 measurement pr.sec. 0001-2 measurements pr.sec. 0010-4 measurements pr.sec. 0011-8 measurements pr.sec. 0100-16 measurements pr.sec. 0101-32 measurements pr.sec. 0110-64 measurements pr.sec. 0111-128 measurements pr.sec. 1000-25/16 measurements pr.sec. 1001-25/8 measurements pr.sec. 1010-25/4 measurements pr.sec. 1011-25/2 measurements pr.sec. 1100-25/1 measurements pr.sec. 1101-50 measurements pr.sec. 1110-100 measurements pr.sec. 1111-200 measurements pr.sec. Applicable for measurements in Background mode only.
TMP_PRC[3:0]	3:0	rw	Temperature oversampling (precision): 0000-single (Default). 0001-2 times. 0010-4 times. 0011-8 times. 0100-16 times. 0101-32 times. 0110-64 times. 0111-128 times. 1xxx-Reserved. Note: temperature measurement time is same as pressure measurement time, refer to Table 14.

6.5 Sensor Operating Mode and Status (MEAS_CFG)

Setup measurement mode

MEAS_CFG			Address:			08H
Measurement configuration			Reset value:			00H
7	6	5	4	3	2	1 0
COEF_RDY	SENSOR_RDY	TEM_RDY	PRS_RDY	TMP_EX T	MEAS_CTRL[2:0]	
r	r	r	r	rw	rw	

Field	Bits	Type	Description
COEF_RDY	7	r	Coefficients will be read to the Coefficients Registers after start-up: 0-Coefficients are not available yet. 1-Coefficients are available.
SENSOR_RDY	6	r	The pressure sensor is running through self-initialization after Start-up. 0-Sensor initialization not complete. 1-Sensor initialization complete. It is recommended not to start measurements until the sensor has completed the self-initialization.
TMP_RDY	5	r	Temperature measurement ready. 1-New temperature measurement is ready. Cleared when temperature measurement is read.
PRS_RDY	4	r	Pressure measurement ready. 1-New pressure measurement is ready. Cleared when pressure measurement is read.
TMP_EXT	3	rw	0-Internal temperature sensor (of ASIC) recommended. 1-External temperature sensor (of MEMS if any).
MEAS_CTRL	2:0	rw	Set measurement mode and type: Standby Mode 000-Idle/Stop background measurement. Command Mode 001-Pressure measurement. 010-Temperature measurement. 011-na. 100-na. Background Mode 101-Continuous pressure measurement. 110-Continuous temperature measurement. 111-Continuous pressure and temperature measurement.

6.6 FIFO configuration (CFG_REG)

Configuration of interrupts, measurement data shift, and FIFO enable.

CFG_REG Address: 09H

Configuration register Reset value: 00H

7	6	5	4	3	2	1	0
INT_HL	INT_FIFO	INT_TMP	INT_PRS	T_SHIFT	P_SHIFT	FIFO_EN	SPI_MODE
rw	rw	rw	rw	rw	rw	rw	rw

Field	Bits	Type	Description
INT_HL	7	rw	Interrupt (on SDO pin) active level: 0-Active low. 1-Active high.
INT_FIFO	6	rw	Generate interrupt when the FIFO is full: 0-Disable. 1-Enable.
INT_TMP	5	rw	Generate interrupt when a temperature measurement is ready: 0-Disable. 1-Enable.
INT_PRS	4	rw	Generate interrupt when a pressure measurement is ready: 0-Disable. 1-Enable.
T_SHIFT	3	rw	Temperature result bit-shift 0-no shift. 1-Shift result right in data register. Note: Must be set to '1' when the oversampling rate is > 8 times.
P_SHIFT	2	rw	Pressure result bit-shift 0-no shift. 1-Shift result right in data register Note: Must be set to '1' when the oversampling rate is > 8 times.
FIFO_EN	1	rw	Enable the FIFO: 0-Disable. 1-Enable.
SPI_MODE	0	rw	Set SPI mode: 0-4-wire interface. 1-3-wire interface.

6.7 Interrupt Status (INT_STS)

Interrupt status register. The register is cleared on read.

INT_STS Address: 0AH

Interrupt status Reset value: 00H

7	6	5	4	3	2	1	0
-	-	-	-	-	INT_FIFO_FULL	INT_TMP	INT_PRS
					r	r	r

Field	Bits	Type	Description
-	7:3	-	Reserved.
INT_FIFO_FULL	2	r	Status of FIFO interrupt 0 - Interrupt not active 1 - Interrupt active
INT_TMP	1	r	Status of temperature measurement interrupt 0 - Interrupt not active 1 - Interrupt active
INT_PRS	0	r	Status of pressure measurement interrupt 0 - Interrupt not active 1 - Interrupt active

6.8 FIFO Status (FIFO_STS)

FIFO status register

FIFO_STS Address: 0BH

FIFO status register Reset value: 00H

7	6	5	4	3	2	1	0
-	-	-	-	-	-	FIFO_FULL	FIFL_EMPTY
						r	r

Field	Bits	Type	Description
-	7:2	-	Reserved.
FIFO_FULL	1	r	0-The FIFO is not full 1-The FIFO is full
FIFO_EMPTY	0	r	0-The FIFO is not empty 1-The FIFO is empty

6.9 Soft Reset and FIFO flush (RESET)

Flush FIFO or generate soft reset.

RESET Address: 0CH

FIFO flush and soft reset Reset value: 00H

7	6	5	4	3	2	1	0
FIFO_FLUSH	-			SOFT_RST			
w	-			w			

Field	Bits	Type	Description
FIFO_FLUSH	7	w	FIFO flush 1-EmptyFIFO After reading out all data from the FIFO, write '1' to clear all old data.
-	6:4	-	Reserved.
SOFT_RST	3:0	w	Write '1001' to generate a soft reset. A soft reset will run through the same sequences as in power-on reset.

6.10 Calibration Coefficients (COEF)

The Calibration Coefficients register contains the 2's complement coefficients that are used to calculate the compensated pressure and temperature values.

Table16 Calibration Coefficients

Coefficient	Addr.	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
c0	0x10	c0[11:4]							
c0/c1	0x11	c0[3:0]				c1[11:8]			
c1	0x12	c1[7:0]							
c00	0x13	c00[19:12]							
c00	0x14	c00[11:4]							
c00/c10	0x15	c00[3:0]				c10[19:16]			
c10	0x16	c10[15:8]							
c10	0x17	c10[7:0]							
c01	0x18	c01[15:8]							
c01	0x19	c01[7:0]							
c11	0x1A	c11[15:8]							
c11	0x1B	c11[7:0]							
c20	0x1C	c20[15:8]							
c20	0x1D	c20[7:0]							
c21	0x1E	c21[15:8]							
c21	0x1F	c21[7:0]							
c30	0x20	c30[15:8]							
c30	0x21	c30[7:0]							

Note: Generate the decimal numbers out of the calibration coefficients registers data:

```
C20 = reg0x1D+reg0x1C*2^8
```

```
if(C20>(2^15-1))
```

```
C20 = C20 - 2^16
```

```
endif
```

```
C0 = (reg0x10*2^4)+((reg0x11/2^4)&0x0F)
```

```
if(C0>(2^11-1))
```

```
C0 = C0-2^12
```

```
endif
```

6.11 Coefficient Source

States which internal temperature sensor the calibration coefficients are based on: the ASIC temperature sensor or the MEMS element temperature sensor. The coefficients are only valid for one sensor and it is highly recommended to use the same temperature sensor in the application. This is set-up in the Temperature Configuration register.

TMP_COEF_SRCE

Address:

28H

Temperature Coefficients Source

Reset value:

xxH

7 6 5 4 3 2 1 0

TMP_COEF_SRCE

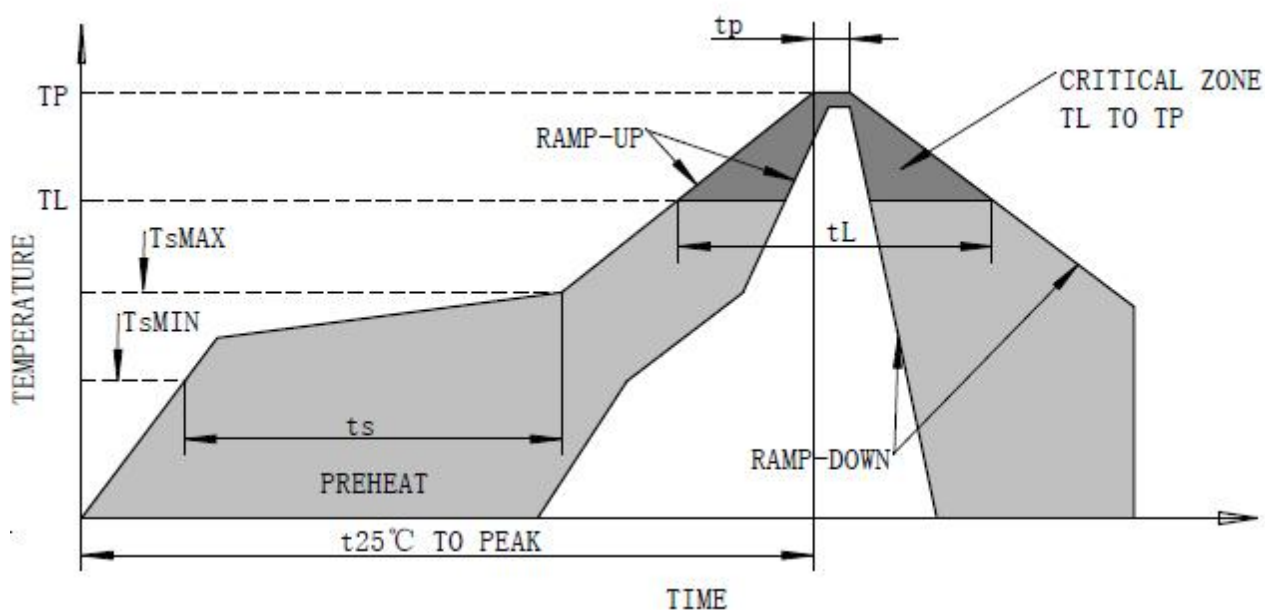
-

r

-

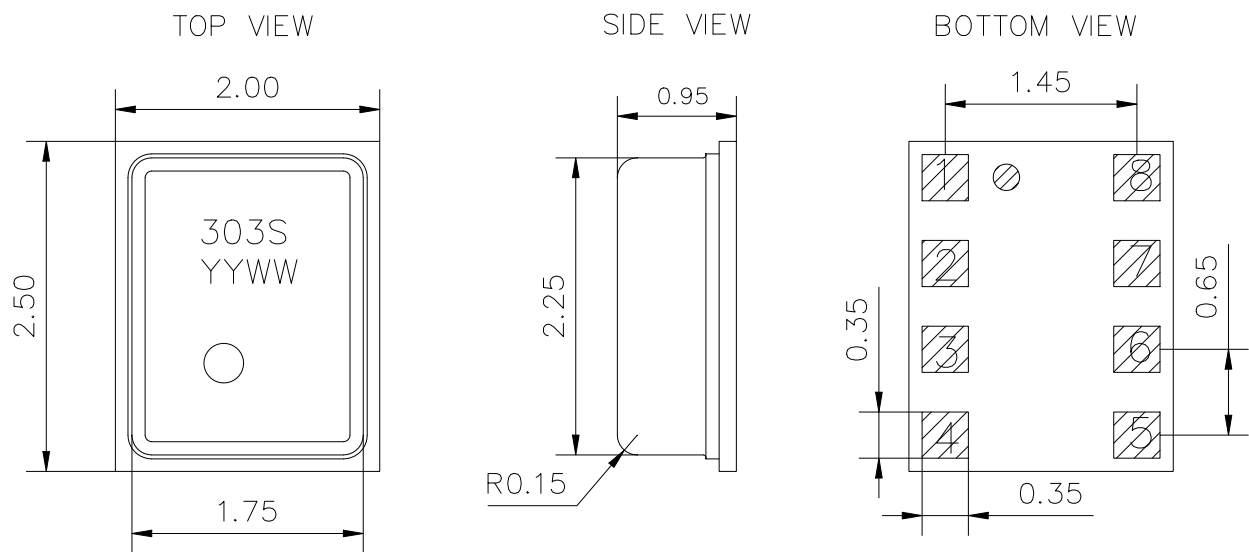
Field	Bits	Type	Description
TMP_COEF_SRCE	7	r	Temperature coefficients are based on: 0-Internal temperature sensor(TBD) 1-External temperature sensor (of pressure sensor MEMS element)
-	6:0	-	Reserved

7. Soldering recommendation



Profile Feature	Sn-Pb Eutectic Assembly	Pb-Free Assembly
Average ramp-up rate(TsMAX to TP)	2°C/seconds max	2°C/seconds max
Preheat		
-Temperature Min.(TsMIN)	80°C	150°C
- Temperature Max.(TsMAX)	160°C	180°C
-Time (TsMIN to TsMAX)(Ts)	60~120 seconds	60~120 seconds
Time maintained above:		
-Temperature(TL)	183°C	217°C
-Time(tL)	20~40 seconds	30~90 seconds
Time 25°C to peak temperature	360 seconds max	480 seconds max
Peak temperature(TP)	220~230°C	230~245°C
Ramp-down rate(peak to 217°C)	2~4°C/seconds	2~4°C/seconds

8. Package dimensions



Notes:

- (1) Mechanical Dimension (unit: mm)
- (2) General tolerance (0.10mm)

9. Tape and Reel Specification

HOPE MICROELECTRONICS CO., LTD

深圳市华普微电子有限公司

版本/Versions

V1.0

图纸编号/Drawing Number:

HP303S卷带

TAPE AND REEL INFORMATION

REEL Diameter

Reel Width (W1)

自粘带封装

Notice:

IX
2 : 1

放置方向如图所示

X
2 : 1

QUADRANT ASSIGNMENTS FOR ORIENTATION IN TAPE

Device	SPQ	Reel diameter (mm)	Reel width (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	D0 (mm)	F (mm)	T (mm)	E (mm)	P0 (mm)	P2 (mm)	Pin1 Quad
HP303S	5000	330	13.1	2.16	2.7	1.15	8	12	1.5	5.5	0.3	1.75	4	2	

技术要求:

- 每10个型腔间距的累积公差为±0.2mm;
- 表面摩擦电压<100V, 10*10⁴欧姆<表面电阻<1*10⁹欧姆;
产品使用自粘带封装;
- 所有尺寸符合EIA-481-3国际标准;
- 每盘13寸盘包装产品数量为5000PCS;
- 每13寸盘载带长度20.8M;
- 材料: 黑色;
- 未注公差为±0.2mm;
- 载带弯曲每100mm不超过1mm;
- A0, B0测量以型腔上表面为基准, 其公差为±0.3mm; K0值测量为型腔内底部为基准至载带上表面测量值。

10. Publication history

Version	Date	Description	Author
V1.0	2021.04.27	New release	Daniel
V1.1	2022.04.01	Update package dimension	Daniel
V1.2	2023.06.19	Update tape and reel specification	Daniel
V1.3	2025.08.20	Update register list	Daniel
V1.4	2026.06.29	Update register description and tape and reel specification	Daniel